

MoRAX: Mobility-based Representation Augmentation for Geospatial Foundation Models

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Abstract

Geospatial Foundation Models (GFM) are emerging as a powerful paradigm for learning semantically rich and geographically consistent visual and physical representations. However, their reliance on Earth-observation (EO) data leaves information about human activity largely underrepresented. Human mobility data reveals the functional and relational structure between regions that is missing from EO data, but is often limited only to the city where it is observed, making it challenging to use for transferable urban representation learning. We introduce **MoRAX**, a lightweight framework for augmenting geospatial embeddings with functional structure derived from human mobility. MoRAX preserves the coverage and consistency of a GFM while providing information about the functional connectivity among urban regions, permitting zero-shot deployment in unseen cities with or without available mobility data. Across four target cities spanning two countries, the MoRAX teacher model, which observes mobility, consistently outperforms GFMs and strong urban representation baselines in eight socioeconomic and environmental prediction tasks. Meanwhile, the student model, which never takes mobility data as input, approaches the teacher in performance on most tasks. Transfer results across countries further demonstrate that modulation conditioned on mobility flows provides a general mechanism for grounding geospatial foundations in the human dimension of cities.

CCS Concepts

• **Information systems** → **Location based services; Geographic information systems.**

1 Introduction

Urban region representation learning aims to encode spatial units of a city into low-dimensional vectors that support a wide range of downstream applications, including socioeconomic data estimation, environmental monitoring, and urban planning [2, 21, 34, 41]. A useful region representation should describe not only the physical and environmental properties of a space but also the functional role that the place plays within the broader urban system. Learning such representations is therefore an important step toward general-purpose urban intelligence [4, 46].

Recent advances in geospatial foundation models (GFMs) provide a promising basis for this goal. By pretraining on large-scale Earth observation (EO) data, these models produce geographically extensive, rich representations of the natural and built environment [3, 11]. AlphaEarth Foundations [3], for instance, integrates multi-source temporal and spatially contextualized Earth observations into globally available, fine-grained geospatial embeddings. Meanwhile, remote-sensing foundation models such as RemoteCLIP [22] extract visual cues directly from satellite imagery. Compared with conventional heterogeneous urban data sources such as points of interest, land-use records, and demographic statistics, geospatial foundation representations offer more consistent, broader coverage across cities and are far less dependent on local taxonomies, individual data platforms, or city-specific collection processes [3, 31]. However, by construction, GFMs learn semantics from what a region and its immediate surrounding environment physically look like, ignoring the fundamentally different (and equally important) information embedded in the *relational* semantics of a city which are defined by how people actually move through, use, and connect urban regions [23]. Traditional GFMs thus remain incomplete for providing a holistic understanding of urban regions.

Human mobility, represented by observed flows of people between regions, has repeatedly been shown to reveal the functional and interaction-driven structure of cities that GFMs do not explicitly capture [40, 45, 48]. It is thus important to investigate how human mobility data can complement geospatial foundation representations. However, treating GFMs and human mobility data as symmetric modalities within a conventional fusion architecture risks obscuring their distinct roles and respective strengths. GFMs provide a compact representation of observable place characteristics and have broad geographic availability. Meanwhile, human mobility data describes relational dynamics among places and is only available in a limited and potentially unrepresentative subset of cities, since collecting it requires dedicated data infrastructure [25].

These observations motivate the approach of learning a lightweight, feature-wise modulation conditioned on mobility in order to adapt, rather than reconstruct, the underlying semantic space of an existing GFM. Such modulation captures how the meaning of a geospatially described place changes according to its exchange relationships with the rest of the city, and allows the adaptation mechanism to be a compact, self-contained module that can be

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learned separately from the base representation it acts upon. This separation allows for distilling the teacher’s mobility-induced modulation into a student model that does not depend on mobility data but approximates the same modulation using only widely available, mobility-free signals. This process enables deployment without access to mobility data, extending applicability across the broad geographic coverage of EO-based GFM. The resulting adaptation and distillation framework generalizes across cities as well as across the choice of baseline foundation model, allowing, for example, the usage of either AlphaEarth or RemoteCLIP as the baseline GFM without changing the overall framework.

Based on this perspective, we propose **MoRAX**, a framework that learns how mobility adapts an existing geospatial foundation representation and distills this adaptation capability into a deployable, mobility-free encoder. A *teacher* is trained on cities with observed mobility graphs, learning a lightweight modulation network that reshapes a baseline foundation embedding into a mobility-aware representation. A *student*, trained on the same cities but never taking mobility as input—mobility influences it only through the teacher’s supervision—is distilled to reproduce this modulation using a proxy relational signal constructed from widely available covariates alone. MoRAX consequently supports two complementary transfer regimes. When mobility is available in an unseen city, the pretrained teacher model directly produces mobility-aware region representations without the need to fine-tune the encoder on the target city. When mobility data is unavailable, the pretrained student model infers how mobility would have reshaped each region’s representation, using only geospatial embeddings and pairwise geographic distances. Fig. 1 provides an overview of the MoRAX framework.

Our main contributions are summarized as follows:

- We develop the MoRAX framework to augment EO-based GFMs with functional mobility structure through lightweight feature-wise modulation.
- We propose a teacher-student framework that distills this modulation into a student model, enabling functionally aware region representations in cities where no mobility data exists.
- We test MoRAX in a series of zero-shot evaluation tasks, finding that the mobility-aware teacher consistently outperforms both geospatial foundation representations and urban representation learning baselines, while the mobility-free student remains competitive with the strongest baselines.

2 Related Work

Transferable Urban Region Representation Learning. Existing urban region representation methods draw on a wide range of data sources, including points of interest [12, 17, 33, 45], mobility flows [37, 42, 48], and satellite imagery [19, 24, 32], which are typically integrated through multimodal fusion and alignment then trained and evaluated within the same city. This diversity in data sources complicates fair comparison across methods, while the single-city paradigm obscures the distinction between regularities that transfer across urban systems and correlations tied to a specific city or its data collection process. These representations thus offer

limited capability of generalization to cities with different urban structure and data modality availability [34].

Existing work has explored large scale urban representation pretraining by organizing heterogeneous urban signals around mobility as the principal relational backbone. By learning from a nation-scale mobility graph, MoRA [40] establishes a shared representation space in which human-centered urban knowledge can be accumulated, compared, and reused across locations. Similarly, PDFM [1] models population dynamics from a rich collection of human-related signals over a large scale neighborhood graph. These studies demonstrate the potential of foundation-style pretraining for learning transferable representations of urban systems from human activity and population dynamics. However, their learned foundations tend to have more limited geographic coverage than EO foundation models (PDFM, for instance, covers only postal codes and counties within the United States).

Earth Observation-Based Geospatial Foundation Models.

Benefiting from the growing volume, diversity, and geographic coverage of Earth observation data, recent EO-based geospatial foundation models have learned increasingly general and reusable representations of the Earth’s surface. For example, AlphaEarth Foundations [3] integrates multi-source, temporal, and spatially contextualized Earth observations into globally available geospatial embeddings while related efforts such as Tessera [11] similarly seek to provide broadly applicable representations for downstream geospatial applications. RemoteCLIP [22] aligns remote sensing imagery with language to support transfer across classification, retrieval, and recognition tasks. However, these representations remain primarily grounded in EO signals and may not fully capture the interaction-driven functional relationships that are critical to urban region understanding.

Adaptation of Pretrained Foundation Models. As pretrained foundation models have grown in scale and capability, research across different domains has developed a range of approaches for adapting their representations to new tasks, domains, and information sources without retraining from scratch [15, 16, 20]. These methods parameterize adaptation through a compact set of task- or domain-specific trainable parameters, learned once and then held fixed at inference time. A complementary line of work instead performs adaptation dynamically, generating the adaptation itself as a function of an auxiliary conditioning signal rather than learning a static set of parameters. FiLM [30], for example, generates feature-wise scaling and shifting parameters from a conditioning input to modulate intermediate activations. Similar conditional modulation mechanisms appear in adaptive normalization methods for style transfer and diffusion models, where activations are adjusted according to style, class, text, or timestep information [9, 18]. Despite their broad use in language and vision, conditional adaptation mechanisms remain underexplored in geospatial representation learning. Geospatial foundation models are still typically used as fixed feature extractors, with limited study of how complementary signals such as city-wide relational information can adapt their pretrained representations.

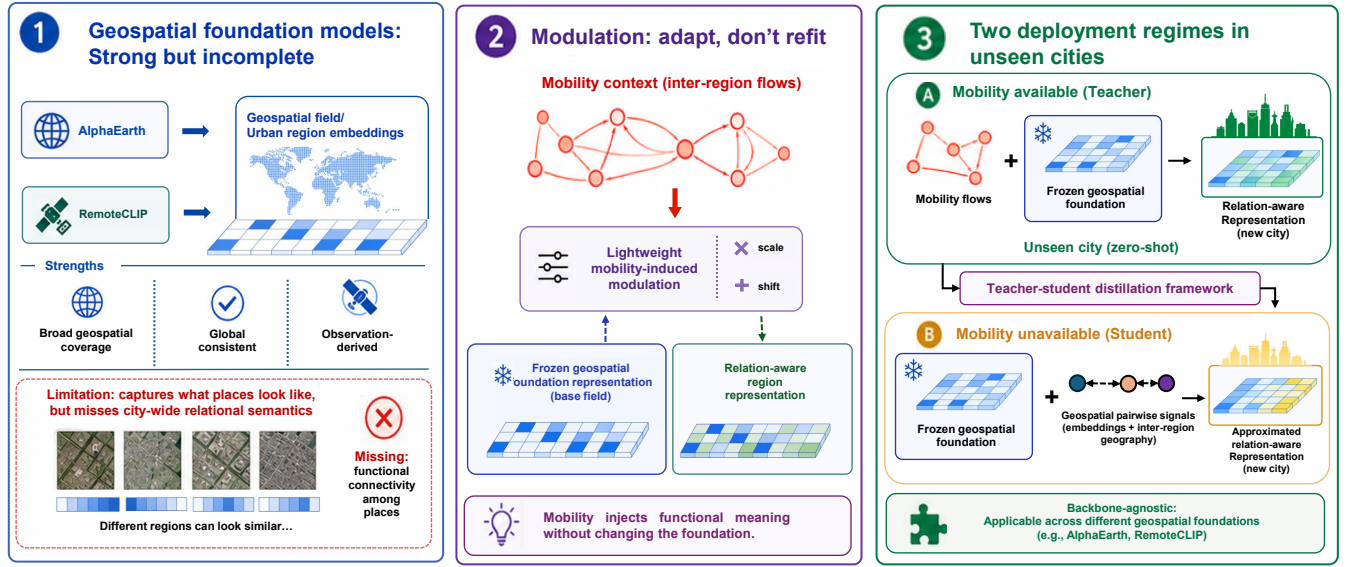


Figure 1: Motivation and overview of MoRAX. Geospatial foundation models (GFM) offer broad, consistent coverage but miss the relational semantics of how places function together (left). MoRAX adapts a frozen GFM base through lightweight modulation induced by mobility (middle), supporting zero-shot deployment in unseen cities with or without available mobility data (right).

3 Preliminaries

Regions. A city c is partitioned into N_c disjoint urban regions, denoted as $\mathcal{R}^c = \{r_1^c, r_2^c, \dots, r_{N_c}^c\}$. For notational simplicity, we omit the city superscript c when the context is clear.

Region Features. Each region is associated with features drawn from different sources, characterizing it from complementary dimensions. We consider two data types, chosen for their independence from manually defined taxonomies (e.g., POI category schemes).

Definition 3.1 (Human Mobility Graph). Human mobility flows record movements between urban regions and reflect functional interactions among them. For a city with observed mobility data, we represent these interactions as a mobility graph $\mathcal{G}_m = (\mathcal{R}, \mathcal{E}_m)$, where each edge $(r_i, r_j) \in \mathcal{E}_m$ is associated with an observed interaction intensity w_{ij} between regions r_i and r_j . The intensity w_{ij} is a generic measure of directed interaction between r_i and r_j within a given period of time, and provides a proxy for the functional connectivity of the two regions. It can be derived, for example, from observed flows of people or trips between the two regions, or from other interaction records aggregated to the region level.

Definition 3.2 (Geospatial Foundation Embedding). Each region r_i is represented by a pretrained foundation embedding s_i , which summarizes the region’s EO-derived physical, environmental, and spatial characteristics in a globally consistent representation space. Such embeddings are produced by encoders pretrained on large-scale EO corpora, including satellite imagery and other sensor-derived measurements, through masked reconstruction, contrastive invariance, and cross-modal alignment objectives [8, 22, 26]. The resulting representation s_i encodes physical morphology such as texture, spectral response, built density, and vegetation, but not the functional role a region plays within the wider urban system.

Region Representation Learning. Given a set of regions $\mathcal{R}$ and their associated features, region representation learning aims to train an encoder $f: \mathcal{R} \rightarrow \mathbb{R}^d$ that maps each region $r_i \in \mathcal{R}$ to a low-dimensional embedding $z_i \in \mathbb{R}^d$ that captures the region r_i ’s characteristics and can be directly applied to a variety of downstream urban prediction tasks with a lightweight model such as Ridge Regression [14].

3.1 Problem Statement

We consider the problem of augmenting pretrained geospatial foundation representations with the city-wide functional structure encoded by human mobility. Given source cities $\mathcal{C}_{\text{src}}$ with fixed foundation embeddings and observed mobility graphs, the objective is to learn a region representation model that transfers zero-shot to unseen target cities $\mathcal{C}_{\text{tgt}}$, without updating the underlying foundation model or performing target-city training; zero-shot thus refers to the absence of target-city parameter updates, not the absence of target-city inputs. At inference time, the target city’s mobility graph may be available or unavailable—in the latter setting, we use a student model that only requires pretrained foundation embeddings and inter-region geographic information.

4 Methodology

4.1 Framework Overview

MoRAX extends pretrained geospatial foundation representations with the city-wide functional structure revealed by human mobility. Its central operation is a lightweight mobility-induced modulation that uses relational context to reinterpret the pretrained representation without discarding its original geospatial semantics.

When an observed mobility graph is available, a mobility-aware teacher derives the conditioning context directly from inter-region flows. When mobility is unavailable, a graph-free student approximates the teacher-induced modulation using only pretrained geospatial embeddings and inter-region geographic information.

As illustrated in Fig. 2, MoRAX first projects the pretrained GFM embedding of each region into a shared urban representation space (Sec. 4.2). It then defines a general feature-wise modulation operator that adapts the projected foundation representation according to a region-level relational context (Sec. 4.3). The teacher obtains this context from observed mobility relations through an edge-centric mobility encoder (Sec. 4.4), whereas the student infers a proxy context from pairwise geospatial representations (Sec. 4.5). Finally, modulation distillation transfers the mobility-induced adaptation from the teacher to the student (Sec. 4.6).

4.2 Geospatial Foundation Representation

As shown in Fig. 2(a), for each region r_i , we retrieve a pretrained GFM embedding $\mathbf{s}_i \in \mathbb{R}^{d_{\text{EO}}}$ according to its geographic location and spatial extent. These embeddings summarize EO-derived characteristics in a globally consistent representation space shared across source and target cities.

We map each pretrained embedding into the hidden space of MoRAX using a lightweight projector

$$\mathbf{z}_i^0 = f_{\text{proj}}(\mathbf{s}_i) \in \mathbb{R}^d, \quad (1)$$

where f_{proj} is implemented as an MLP shared across all regions and cities. The underlying geospatial foundation representation remains frozen and the projector is shared between the student and the teacher.

The resulting set $\mathbf{Z}^0 = \{\mathbf{z}_i^0\}_{i=1}^N$ provides the shared geospatial foundation representation used by both the teacher and student. Because it is independent of the mobility graph, it provides a common semantic coordinate system across cities and across mobility availability settings. We refer to this spatially indexed layer of region embeddings, defined consistently within and across cities, as the *geospatial field*. However, the pretrained representation primarily describes the observable characteristics of individual regions and does not explicitly encode how they interact as parts of a city-wide system. We therefore conditionally reshape this representation using region-level relational context.

4.3 Relation-Induced Feature Modulation

We first define the modulation operator, which is independent of how the conditioning signal is obtained. Given the projected foundation embedding $\mathbf{z}_i^0$ and a region-level relational context $\mathbf{z}_i^r$, a lightweight modulation network generates feature-wise scale and shift parameters

$$\boldsymbol{\gamma}_i = f_{\gamma, \theta}(\mathbf{z}_i^r), \quad \boldsymbol{\beta}_i = f_{\beta, \theta}(\mathbf{z}_i^r), \quad (2)$$

where $\boldsymbol{\gamma}_i, \boldsymbol{\beta}_i \in \mathbb{R}^d$, and θ denotes the parameters of the modulation module.

We parameterize the feature-wise scale as $1 + \tanh(\boldsymbol{\gamma}_i)$, which constrains each scaling coefficient to $(0, 2)$. This parameterization centers the scale at identity and allows mobility to induce a bounded, feature-wise reweighting of the projected base. The

resulting relation-conditioned transformation is

$$\mathbf{z}_i^{\text{mod}} = (1 + \tanh(\boldsymbol{\gamma}_i)) \odot \mathbf{z}_i^0 + \boldsymbol{\beta}_i, \quad (3)$$

where $\odot$ denotes element-wise multiplication.

For compactness, we denote the complete operation in Eqs. (2) and (3) as

$$\mathbf{z}_i^{\text{mod}} = \mathcal{M}_{\theta}(\mathbf{z}_i^0, \mathbf{z}_i^r). \quad (4)$$

The modulated features are then integrated with the projected foundation representation

$$\mathbf{z}_i = \mathbf{z}_i^0 + \mathbf{z}_i^{\text{mod}}. \quad (5)$$

This additive formulation retains the projected geospatial representation while allowing relational context to alter its feature-wise interpretation. The projected foundation representation continues to provide the semantic basis of the final representation, while the modulation introduces complementary information about the functional role of each region.

The modulation operation is agnostic to the source of $\mathbf{z}_i^r$. In the mobility-aware teacher, the relational context is encoded directly from observed mobility flows. In the graph-free student, it is inferred from pairwise geospatial representations and inter-region geography. The teacher and student use the same modulation form but have independently learned parameters, denoted by θ_T and θ_S , respectively, with only the frozen foundation embeddings and the projector f_{proj} being shared between the two.

4.4 Teacher: Mobility-Aware Relational Encoding

When inter-region mobility flows are observed, we derive the relational context directly from the mobility graph. We refer to this mobility-aware instantiation as the teacher because its induced modulation later supervises the graph-free student.

Because mobility is naturally defined over region pairs rather than individual regions, we adopt an edge-centric encoder that explicitly updates relation states before aggregating them into region-level mobility contexts, as shown in Fig. 2(b).

4.4.1 Edge-Centric Mobility Encoding. Human mobility is naturally observed on region pairs rather than on individual regions. For each mobility edge $(r_i, r_j) \in \mathcal{E}_m$, let $\mathbf{a}_{ij}^m$ denote its edge attributes, including within-city standardized flow strength and geographic distance. We first map these attributes into a hidden relation state

$$\mathbf{e}_{ij}^{(0)} = f_{\text{edge}}(\mathbf{a}_{ij}^m). \quad (6)$$

The mobility encoder then applies L edge-centric message-passing layers. At layer ℓ , each region summarizes its incident relation states into a relational profile

$$\mathbf{h}_i^{(\ell)} = \text{Agg} \left(\left\{ \mathbf{e}_{ji}^{(\ell-1)} : (r_j, r_i) \in \mathcal{E}_m \right\} \right), \quad (7)$$

where Agg is mean pooling in our implementation, and $\mathbf{h}_i^{(\ell)}$ is set to the zero vector for regions with no incoming edges.

Each edge state is then updated using the relational profiles of both endpoints and its previous state

$$\mathbf{e}_{ij}^{(\ell)} = \mathbf{e}_{ij}^{(\ell-1)} + f_{\text{upd}}^{(\ell)} \left(\mathbf{h}_i^{(\ell)} \parallel \mathbf{h}_j^{(\ell)} \parallel \mathbf{e}_{ij}^{(\ell-1)} \right), \quad (8)$$

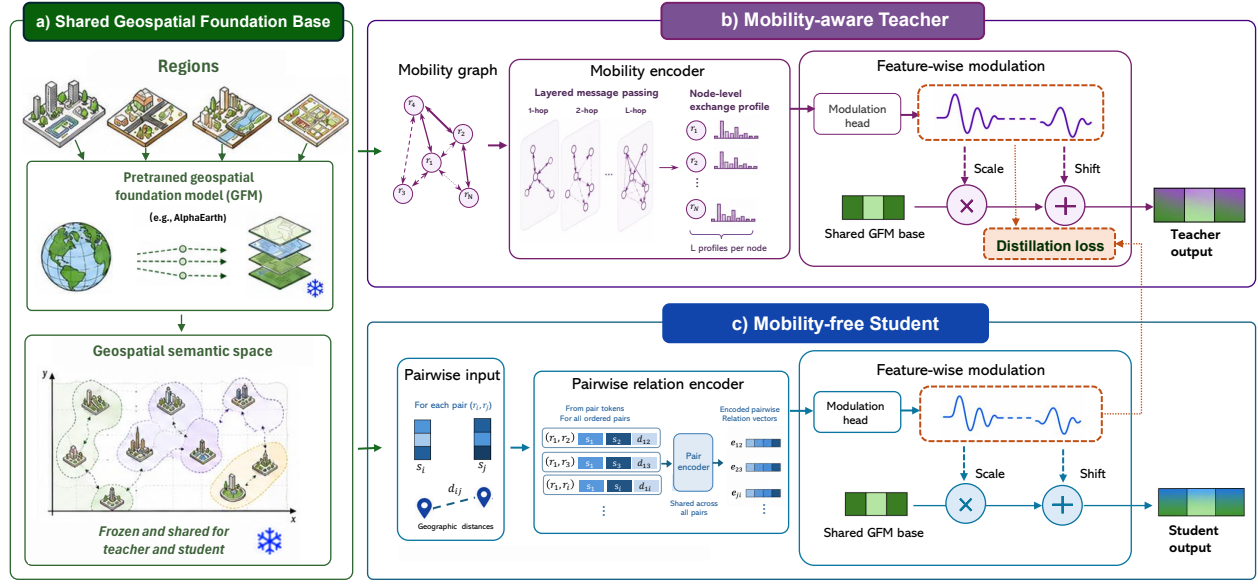


Figure 2: The MoRAX architecture. (a) A frozen geospatial foundation model provides a shared semantic base in which pretrained region embeddings are projected into a common representation space used by both teacher and student. (b) The mobility-aware teacher encodes the observed mobility graph with an edge-centric message passing encoder and generates feature-wise scale and shift parameters that modulate the shared base. (c) The mobility-free student approximates this modulation from pairwise geospatial embeddings and inter-region distances alone, trained by distilling the teacher’s mobility-induced modulation.

where $\parallel$ denotes vector concatenation and $f_{\text{upd}}^{(\ell)}$ is a layer-specific MLP. Since the readout below consumes only the relational profiles, the edge update is omitted at the final layer $\ell = L$.

Finally, the mobility context of region r_i is read out from the relational profiles across all layers

$$\mathbf{z}_i^m = f_{\text{read}}(\mathbf{h}_i^{(1)} \parallel \dots \parallel \mathbf{h}_i^{(L)}) \in \mathbb{R}^d. \quad (9)$$

Since $\mathbf{z}_i^m$ is constructed from explicitly updated relation states, it summarizes the multi-layer relational profile associated with region r_i , rather than merely smoothing neighboring node features. The resulting mobility context serves as the teacher-side relational conditioning signal, $\mathbf{z}_i^{r,T} = \mathbf{z}_i^m$.

The teacher then applies the modulation operator defined in Eq. (4)

$$\mathbf{z}_i^{\text{mod},T} = \mathcal{M}_{\theta_T}(\mathbf{z}_i^0, \mathbf{z}_i^{r,T}), \quad \mathbf{z}_i^T = \mathbf{z}_i^0 + \mathbf{z}_i^{\text{mod},T}. \quad (10)$$

The teacher thus uses observed city-wide interactions to reinterpret the foundation representation while retaining its original geospatial content. Its computation, however, requires access to the target-city mobility graph. We therefore introduce a graph-free student that approximates the teacher-induced modulation from inputs available without mobility observations.

4.5 Student: Graph-Free Modulation Approximation

As shown in Fig. 2(c), the student does not attempt to reconstruct the target mobility graph. Instead, it learns to approximate how

the mobility-aware teacher reshapes each region representation. It operates on the same projected foundation representation as the teacher but infers its relational context from pairwise geospatial embeddings and inter-region geographic information.

4.5.1 Geospatial Pair Relation Encoder. For an anchor region r_i and a candidate partner r_j , where $j \neq i$, we construct the pair attribute

$$\mathbf{a}_{ij}^p = [\mathbf{s}_i \parallel \mathbf{s}_j \parallel d_{ij}], \quad (11)$$

where $\mathbf{s}_i$ and $\mathbf{s}_j$ are the pretrained GFM embeddings of the two regions, and d_{ij} is their geographic distance standardized within the city.

A shared pair encoder maps each pair attribute into a latent relation representation

$$\mathbf{e}_{ij}^S = f_{\text{pair}}(\mathbf{a}_{ij}^p) \in \mathbb{R}^d. \quad (12)$$

For each anchor region, the student aggregates the relation representations of all candidate partners

$$\mathbf{z}_i^p = \frac{1}{N-1} \sum_{j \neq i} \mathbf{e}_{ij}^S, \quad (13)$$

The pair encoder captures the geospatial relation between the anchor and each candidate region, while mean pooling produces a permutation-invariant summary over the city. This aggregated representation serves as the student-side relational conditioning signal, $\mathbf{z}_i^{r,S} = \mathbf{z}_i^p$.

The student applies the same modulation form with independently learned parameters

$$\mathbf{z}_i^{\text{mod},S} = \mathcal{M}_{\theta_S} \left(\mathbf{z}_i^0, \mathbf{z}_i^{r,S} \right), \quad \mathbf{z}_i^S = \mathbf{z}_i^0 + \mathbf{z}_i^{\text{mod},S}. \quad (14)$$

4.6 Training Objectives

We train MoRAX in two stages. The teacher is first optimized on source-city mobility graphs so that observed interaction structure is encoded into both the mobility context and the final adapted representation. The teacher is then frozen, and its mobility-induced modulation supervises the graph-free student.

4.6.1 Teacher Training. The teacher is trained using self-supervision derived from observed mobility relations. The objective is applied both to the mobility context and to the final teacher representation.

Mobility Relation Loss. For each anchor region r_i , we construct a candidate set $C_i = \mathcal{P}_i \cup \mathcal{N}_i$, where $\mathcal{P}_i$ contains all regions with observed mobility interactions from r_i , and $\mathcal{N}_i$ contains distance-stratified sampled non-neighbors.

Positive candidates are assigned flow-normalized soft targets

$$q_{ij} = \frac{w_{ij}}{\sum_{r_k \in \mathcal{P}_i} w_{ik}}, \quad r_j \in \mathcal{P}_i, \quad (15)$$

while negative candidates receive zero target mass $q_{ij} = 0$. Given a region representation $\mathbf{u}_i$, a level-specific pair scorer ψ_u predicts a logit for each candidate

$$s_{ij}^u = \psi_u([\mathbf{u}_i \| \mathbf{u}_j]), \quad p_{ij}^u = \frac{\exp(s_{ij}^u)}{\sum_{r_k \in C_i} \exp(s_{ik}^u)}. \quad (16)$$

The mobility relation loss is the soft-target cross entropy

$$\mathcal{L}_{\text{mrel}}(\mathbf{u}) = -\frac{1}{|\mathcal{A}|} \sum_{r_i \in \mathcal{A}} \sum_{r_j \in C_i} q_{ij} \log p_{ij}^u, \quad (17)$$

where $\mathcal{A}$ denotes anchors with at least one observed positive interaction.

We apply this objective at two levels. First, $\mathcal{L}_{\text{mrel}}^m = \mathcal{L}_{\text{mrel}}(\mathbf{z}^m)$ directly supervises the mobility encoder. Second, $\mathcal{L}_{\text{mrel}}^T = \mathcal{L}_{\text{mrel}}(g_T(\mathbf{z}^T))$ applies the same supervision to a relation-head projection of the final teacher representation, where g_T is a lightweight MLP head, to preserve mobility-derived interaction structure.

The overall teacher objective is then as follows

$$\mathcal{L}_T = \mathcal{L}_{\text{mrel}}^m + \lambda_T \mathcal{L}_{\text{mrel}}^T, \quad (18)$$

where λ_T controls the contribution of the representation-level relation objective. Note that the teacher objective is purely relational, since the underlying geospatial semantics are preserved architecturally, through the frozen foundation embeddings and the additive formulation of Eq. (5), rather than through an explicit grounding term as in the student objective below.

4.6.2 Student Training. The student learns to approximate the mobility-conditioned modulation generated by the teacher.

Distillation of relation-induced modulation. We align the student modulated features with the detached teacher features

$$\mathcal{L}_{\text{mod}} = \frac{1}{N} \sum_{i=1}^N \left\| \mathbf{z}_i^{\text{mod},S} - \mathbf{z}_i^{\text{mod},T} \right\|_2^2. \quad (19)$$

To keep the student representation anchored to the pretrained GFM semantic space, we use a lightweight decoder Dec_S to reconstruct the original foundation embedding

$$\mathcal{L}_{\text{field}}^S = \frac{1}{N} \sum_{i=1}^N \left\| \text{Dec}_S(\mathbf{z}_i^S) - \mathbf{s}_i \right\|_2^2. \quad (20)$$

The overall student objective is

$$\mathcal{L}_S = \alpha_{\text{mod}} \mathcal{L}_{\text{mod}} + \mathcal{L}_{\text{field}}^S, \quad (21)$$

where α_{mod} controls the weight of the modulation distillation term.

5 Experiments

5.1 Experimental Setup

Datasets. We collect data from seven Chinese cities. Five cities, Beijing, Shenzhen, Hangzhou, Nanjing, and Suzhou, serve as source cities for pretraining, while **Shanghai (SH)** and **Guangzhou (GZ)** are held out for zero-shot transfer evaluation. We further include **New York City (NYC)** and **Chicago (CHI)** to assess cross-country transfer. We use AlphaEarth embeddings as the EO-based geospatial foundation representation in the main experiments. We adopt hexagonal H3 grids [35] at Level 7 for region discretization for Shanghai and Guangzhou, with each cell covering approximately 5.16 km², offering a spatial granularity suitable for city-level modeling. For NYC and CHI, administrative boundaries (i.e., census tracts) are adopted for region construction. We detail the training and downstream data collection and processing in Appendix C.

Implementation Details. MoRAX is trained in two stages. We first train the teacher on the 5 source cities with observed mobility graphs, and then freeze the teacher to train the student on the same source cities. Both stages use AdamW, with learning rates 3×10^{-4} and 1×10^{-4} for the teacher and student, respectively, and weight decay 1×10^{-2} . The hidden dimension is $d = 128$, the mobility encoder uses $L = 3$ message-passing layers, and the loss weights are $\lambda_T = 0.5$ and $\alpha_{\text{mod}} = 0.25$. For each anchor region, $\mathcal{N}_i$ contains up to 24 negatives sampled with distance stratification: non-neighbor regions are divided into three distance-quantile bins according to their distance to r_i , and up to eight negatives are sampled from each bin.

Evaluation Procedure. We evaluate two deployment regimes, corresponding to the teacher and student encoders introduced in Sec. 4. In the *mobility-available* regime, the pretrained teacher f_T is directly applied to a held-out target city using that city's own mobility graph, without any target-city fine-tuning. In the *mobility-free* regime, the pretrained student f_S is applied to the same held-out city using only widely available region signals, without observing the target city's mobility graph.

We consider four downstream prediction tasks for Shanghai and Guangzhou: crime, nighttime light intensity, carbon emissions, and PM_{2.5} concentration. For New York City and Chicago, we evaluate crime, check-in counts, house prices, and income levels. For each downstream task, we train a lightweight Ridge regressor on the target city and evaluate it with five-fold cross-validation over regions. We report the coefficient of determination, R^2 , as the main metric.

Baselines. We select state-of-the-art urban region representation methods that are compatible with city-agnostic transfer as baselines, including HREP [48], FlexiReg [32], and UrbanCLIP [43].

Table 1: Comparison of model performance on cross-city and cross-country transfer. Results are reported as R^2 . The best result for each task is shown in bold and the second best is underlined. Δ denotes the relative improvement over the strongest non-MoRAX baseline for each task. Standard deviations over 10 random seeds are reported in parentheses (RemoteCLIP and AlphaEarth embeddings are deterministic and therefore reported without variance).

Model	Shanghai				Guangzhou			
	Crime	NTL	CO ₂	PM _{2.5}	Crime	NTL	CO ₂	PM _{2.5}
RemoteCLIP	0.494	0.636	0.449	0.397	0.524	0.611	0.433	0.346
AlphaEarth	0.548	0.638	0.508	0.506	0.672	0.694	0.466	0.430
FlexiReg	0.192(0.037)	0.237(0.040)	0.228(0.038)	0.281(0.050)	0.209(0.052)	0.225(0.060)	0.257(0.058)	0.399(0.079)
HREP	0.268(0.014)	0.361(0.027)	0.323(0.006)	0.592(0.044)	0.367(0.025)	0.415(0.036)	0.355(0.027)	0.640(0.035)
UrbanCLIP	0.519(0.007)	0.642(0.004)	0.418(0.001)	0.287(0.017)	0.563(0.008)	0.633(0.008)	0.369(0.024)	0.281(0.019)
MoRAX-Student	0.569(0.008)	0.774(0.002)	0.659(0.002)	0.915(0.001)	0.725(0.017)	0.797(0.018)	0.702(0.012)	0.929(0.005)
Student Δ	+3.8%	+20.6%	+29.7%	+54.6%	+7.9%	+14.8%	+50.6%	+45.2%
MoRAX-Teacher	0.618(0.005)	0.777(0.010)	0.649(0.004)	0.917(0.002)	0.727(0.005)	0.803(0.007)	0.696(0.004)	0.943(0.002)
Teacher Δ	+12.8%	+21.0%	+27.8%	+54.9%	+8.2%	+15.7%	+49.4%	+47.3%

Model	NYC				Chicago			
	Crime	Check-in	House	Income	Crime	Check-in	House	Income
RemoteCLIP	0.168	0.341	0.232	0.271	0.012	0.243	0.188	0.113
AlphaEarth	0.179	0.464	0.222	0.277	0.050	0.246	0.208	0.108
FlexiReg	0.176(0.043)	0.256(0.078)	0.179(0.055)	0.209(0.044)	0.174(0.086)	0.233(0.061)	0.307(0.093)	0.235(0.100)
HREP	0.421(0.039)	0.564(0.042)	0.419(0.049)	0.518(0.027)	0.382(0.052)	0.610(0.053)	0.625(0.047)	0.455(0.060)
UrbanCLIP	0.150(0.033)	0.318(0.038)	0.111(0.018)	0.149(0.026)	0.066(0.021)	0.205(0.054)	0.163(0.048)	0.106(0.037)
MoRAX-Student	0.404(0.013)	0.588(0.013)	0.466(0.034)	0.530(0.022)	0.367(0.030)	0.570(0.040)	0.715(0.014)	0.502(0.024)
Student Δ	-4.0%	+4.3%	+11.2%	+2.3%	-3.9%	-6.6%	+14.4%	+10.3%
MoRAX-Teacher	0.520(0.035)	0.825(0.020)	0.469(0.064)	0.542(0.039)	0.662(0.050)	0.859(0.021)	0.809(0.051)	0.622(0.051)
Teacher Δ	+23.5%	+46.3%	+11.9%	+4.6%	+73.3%	+40.8%	+29.4%	+36.7%

For each method, we train on the five source cities and directly apply the learned model to the four test cities, reporting the average transfer performance across source-city runs.

We further include two pretrained geospatial foundation models as training-free baselines. **RemoteCLIP** [22] directly encodes satellite imagery from the target cities, whereas **AlphaEarth** [3] provides a pretrained geospatial embedding for each target region, retrieved according to its geographic location and spatial extent.

5.2 Performance Comparison

Table 1 reports cross-city transfer results on four held-out cities, covering both within-country and cross-country settings. Overall, MoRAX-Teacher consistently outperforms pretrained foundation embeddings and existing urban representation baselines, achieving relative gains of up to 54.9% in the within-country setting and 73.3% in the cross-country setting.

Pretrained geospatial embeddings, particularly AlphaEarth, perform strongly in the Chinese target cities, where tasks such as CO₂ prediction are closely associated with spatially continuous physical and environmental patterns that Earth-observation-based foundations can capture. Their advantage is less consistent in the U.S. cities, where the prediction targets depend more strongly on human activity and inter-region interactions and are therefore less

directly recoverable from Earth-observation signals alone. This contrast highlights the need to complement Earth-observation-based foundation embeddings with relational knowledge.

Built on frozen AlphaEarth embeddings, MoRAX-Teacher increases the average R^2 from 0.558 to 0.766 in the within-country setting and from 0.219 to 0.663 in the cross-country setting, corresponding to relative gains of 37.4% and 202.6%, respectively. Importantly, these gains are not confined to tasks for which the underlying foundation representation is weak. Even when AlphaEarth already performs strongly, as in NTL prediction for Shanghai and Guangzhou, relational adaptation yields further improvements. This suggests that mobility contributes complementary functional structure rather than merely compensating for deficiencies in the base representation. Overall, mobility conditioned modulation enables MoRAX to enrich geospatial foundation representations with interaction-aware information while preserving their observation-derived semantics.

MoRAX-Student further shows that a substantial portion of the mobility-induced modulation can be transferred to a graph-free encoder through distillation. On spatially smooth tasks, the student closely matches the teacher and occasionally performs slightly better. In contrast, the larger teacher-student gaps on crime and

check-in prediction suggest that some fine-grained human interaction patterns remain difficult to recover without direct access to the target city’s mobility graph.

Nevertheless, MoRAX-Student improves over AlphaEarth by an average of 36.0% in the within-country setting and 136.1% in the cross-country setting. It also outperforms most urban region representation baselines despite its lightweight, graph-free design. The main exception is HREP on several interaction-intensive U.S. tasks, where human mobility is used directly as an input.

5.3 Cross-Country Transfer

We further complement the cross-country evaluation with a bidirectional transfer analysis. In addition to transferring models trained on the five Chinese source cities to New York City and Chicago, we reverse the direction by training MoRAX on the two U.S. cities and directly applying it to the held-out Chinese cities. As shown

Table 2: Cross-country transfer results (R^2).

(a) Transfer to held-out Chinese cities							
Target	Source	Encoder	Crime	NTL	CO ₂	PM _{2.5}	
SH	–	AlphaEarth	0.548	0.638	0.508	0.506	
		Student	0.569	0.774	0.659	0.915	
	5 CN	Teacher	0.618	0.777	0.649	0.917	
		NYC+CHI	Student	0.585	0.789	0.615	0.829
			Teacher	0.625	0.785	0.597	0.824
GZ	–	AlphaEarth	0.672	0.694	0.466	0.430	
		Student	0.725	0.797	0.702	0.929	
	5 CN	Teacher	0.727	0.803	0.696	0.943	
		NYC+CHI	Student	0.736	0.843	0.565	0.818
			Teacher	0.742	0.849	0.541	0.782

(b) Transfer to U.S. cities							
Target	Source	Encoder	Crime	Check-in	House	Income	
NYC	–	AlphaEarth	0.179	0.464	0.222	0.277	
		Student	0.404	0.588	0.466	0.530	
	5 CN	Teacher	0.520	0.825	0.469	0.542	
		CHI	Student	0.349	0.537	0.402	0.536
			Teacher	0.422	0.752	0.410	0.541
CHI	–	AlphaEarth	0.050	0.246	0.208	0.108	
		Student	0.367	0.570	0.715	0.502	
	5 CN	Teacher	0.662	0.859	0.809	0.622	
		NYC	Student	0.453	0.517	0.651	0.425
			Teacher	0.476	0.856	0.700	0.525

in Table 2(a), training on the five Chinese cities or on the two U.S. cities leads to broadly comparable performance in Shanghai and Guangzhou. More importantly, both source configurations yield consistent improvements over the unadapted AlphaEarth embeddings across all tasks for both the student and teacher. This suggests that mobility-conditioned modulation captures relational regularities that can transfer across countries, rather than merely memorizing interaction patterns specific to the source cities.

Interestingly, for transfer to New York City and Chicago, training on the five Chinese cities generally performs better than training on the other U.S. city (See Table 2 (b)), despite the latter being geographically and institutionally closer to the target. Averaged across the student and teacher variants, the five-city Chinese source

Table 3: Performance applying MoRAX to RemoteCLIP (R^2). Δ denotes the relative improvement over RemoteCLIP.

City	Model	Crime	NTL	CO ₂	PM _{2.5}
SH	RemoteCLIP	0.494	0.636	0.449	0.397
	MoRAX-Student	0.532	0.716	0.632	0.874
	Student Δ	+7.7%	+12.6%	+40.8%	+120.2%
	MoRAX-Teacher	0.596	0.739	0.623	0.884
	Teacher Δ	+20.6%	+16.2%	+38.8%	+122.7%
GZ	RemoteCLIP	0.524	0.611	0.433	0.346
	MoRAX-Student	0.655	0.748	0.687	0.895
	Student Δ	+25.0%	+22.4%	+58.7%	+158.7%
	MoRAX-Teacher	0.702	0.801	0.635	0.931
	Teacher Δ	+34.0%	+31.1%	+46.7%	+169.1%

set improves the mean R^2 from 0.535 to 0.591, corresponding to a relative gain of 10.4%, and achieves higher performance in 14 of the 16 model–task comparisons. This result suggests that source diversity may be more important than geographic proximity alone. The five Chinese cities expose the model to a broader range of urban forms and interaction patterns, which may support a more robust and transferable modulation rule, whereas training on a single source city is more susceptible to city-specific regularities.

5.4 Generalization Across Geospatial Foundations

Table 3 evaluates whether MoRAX generalizes beyond AlphaEarth by applying the same adaptation framework to RemoteCLIP, a satellite-based vision-language foundation model. Across Shanghai and Guangzhou, both MoRAX variants consistently improve the original RemoteCLIP representations on all four downstream tasks.

To accommodate the independently encoded region images produced by RemoteCLIP, we first transform them into a spatially continuous visual field using parameter-free heat diffusion, as described in Appendix A. We then apply the same relational modulation framework used for the AlphaEarth-derived geospatial field. The consistent gains across these two substantially different foundation representations suggest that MoRAX provides a general interface for relational adaptation across geospatial backbones. Its improvements arise from injecting city-wide interaction structure into a spatially aligned region field, rather than from properties specific to the underlying geospatial encoder. Additional ablations in Appendix B further show that constructing a coherent semantic field is important for effective modulation.

5.5 Ablation Study

We conduct ablation studies on Shanghai to isolate the contribution of each component in MoRAX. For the teacher, **w/o modulation** directly uses the original AlphaEarth embedding without mobility-induced reshaping; **w/o relation** removes mobility-relation supervision and replaces it with geospatial reconstruction; **w/o foundation** removes the geospatial foundation and trains only the mobility encoder. For the student, **w/o distillation** trains the student directly from observed mobility relations without teacher supervision while **w/o modulation alignment** removes the objective aligning the student with the teacher’s mobility-induced modulation.

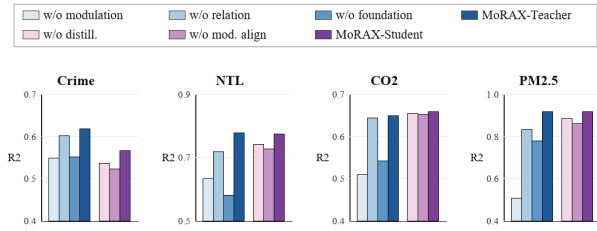


Figure 3: Ablation study on Shanghai (R^2). Blue bars ablate the teacher by removing the modulation, the mobility-relation supervision, or the geospatial foundation. Purple bars ablate the student by training directly on mobility relations without the teacher (w/o distill.) or removing modulation alignment (w/o mod. align). The complete teacher and student match or exceed every ablated variant on all four tasks.

As shown in Figure 3, the gap between the mobility-only variant (w/o foundation) and the teacher indicates that mobility provides useful relational information but benefits from the stable semantic base of the geospatial foundation. Removing relation supervision also degrades the teacher, showing that reconstructing the original AlphaEarth embeddings alone is insufficient to learn meaningful modulation. For the student, both direct training and removing modulation alignment reduce performance on most tasks, suggesting that teacher supervision helps transfer the mobility-induced adaptation rather than only providing another embedding target. Overall, the ablation results support the central design of MoRAX: a shared geospatial foundation provides transferable semantics, mobility-relation learning injects functional context through feature-wise modulation, and distillation transfers this adaptation to the graph-free student.

6 Conclusion

We presented MoRAX, a framework that learns transferable urban region representations by using mobility to modulate a shared geospatial foundation. The key idea is that foundation embeddings provide a stable cross-city semantic base, but Earth-observation alone is not sufficient for urban understanding. MoRAX therefore treats mobility as relational context that reshapes the functional interpretation of geospatial foundation embeddings.

MoRAX further enables graph-free transfer by distilling the modulation process itself into a student that relies only on widely available proxy signals (e.g. pairwise distances). Across Chinese and U.S. cities, MoRAX consistently improves over pretrained geospatial embeddings and existing urban representation baselines. It further demonstrates the generalizability across foundation model backbones and across countries, strengthening the foundation backbone along a human-centric dimension. More broadly, this work suggests a shift in how relational signals can be incorporated into the growing ecosystem of pretrained foundation models.

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A Constructing a Visual Field from RemoteCLIP

Unlike AlphaEarth, which directly provides a spatially contextualized geospatial embedding field, RemoteCLIP independently encodes the satellite image associated with each urban region. Directly applying relational modulation to these isolated image embeddings provides no explicit spatial continuity across neighboring regions. We therefore apply a parameter-free heat diffusion operator before using RemoteCLIP as the field base of MoRAX.

For each region r_i , a pretrained RemoteCLIP encoder produces an image embedding $\mathbf{s}_i \in \mathbb{R}^{d_{RC}}$. Let $\mathbf{S} \in \mathbb{R}^{N \times d_{RC}}$ denote the embedding matrix of all regions in a city. We construct a distance-weighted K -nearest-neighbor graph ($K = 8$) over region centroids. The edge weight between neighboring regions is defined by an exponential decay of geographic distance. The graph is then symmetrized and row-normalized to obtain a transition matrix $\mathbf{P}$.

Let $\mathcal{H} = \{0, 1, 2, 4\}$ denote the retained diffusion orders. The spatially diffused embedding of region r_i is

$$\tilde{\mathbf{s}}_i = \sum_{h \in \mathcal{H}} \mathbf{w}_h(t) \left(\mathbf{P}^h \mathbf{S} \right)_i, \quad (22)$$

where the diffusion weights follow a truncated and renormalized Poisson distribution with diffusion time $t = 2$:

$$\mathbf{w}_h(t) = \frac{e^{-t} t^h / h!}{\sum_{h' \in \mathcal{H}} e^{-t} t^{h'} / h'!}. \quad (23)$$

We then project the diffused satellite embedding into the model hidden space as $\mathbf{z}_i^0 = f_{\text{proj}}(\tilde{\mathbf{s}}_i) \in \mathbb{R}^d$. The resulting $\mathbf{z}_i^0$ serves as the RemoteCLIP-based visual field for both the teacher and student. Heat diffusion is used only to construct a spatially coherent field from independently encoded region images, while the subsequent mobility encoder, pairwise student encoder, modulation heads, and training objectives remain unchanged.

B Importance of the Semantic Field

MoRAX assumes that relational context modulates a coherent region-level semantic field. To examine the importance of this field structure, we compare the complete RemoteCLIP-based model with a variant that applies the same teacher and student modulation mechanisms directly to isolated RemoteCLIP region embeddings, without heat diffusion. All other components and training settings are kept unchanged.

Table 4: Ablation on the RemoteCLIP-based semantic field. “w/ Field” applies heat diffusion before projection, whereas “w/o Field” directly projects isolated RemoteCLIP embeddings. Results are reported in terms of R^2 .

Task	Variant	Target Cities	
		GZ	SH
Crime	Student w/o Field	0.415	0.432
	Student w/ Field	0.655	0.532
	Teacher w/o Field	0.571	0.534
	Teacher w/ Field	0.702	0.596
NTL	Student w/o Field	0.572	0.628
	Student w/ Field	0.748	0.716
	Teacher w/o Field	0.651	0.639
	Teacher w/ Field	0.801	0.739
CO ₂	Student w/o Field	0.237	0.449
	Student w/ Field	0.687	0.632
	Teacher w/o Field	0.267	0.491
	Teacher w/ Field	0.635	0.623
PM _{2.5}	Student w/o Field	0.345	0.384
	Student w/ Field	0.895	0.874
	Teacher w/o Field	0.776	0.690
	Teacher w/ Field	0.931	0.884

As reported in Table 4, removing heat diffusion consistently and largely weakens both the teacher and student. This result indicates that relational modulation benefits from an aligned and spatially coherent base: the mobility or proxy context can then reinterpret an existing semantic field rather than independently modulating a set of disconnected image embeddings.

C Datasets

C.1 Training Datasets

Mobility graphs. For U.S. cities, we use publicly available ride-hailing trip records released by local transportation agencies. New York City mobility flows are constructed from the Taxi and Limousine Commission (TLC) trip record data [7], while Chicago mobility flows are constructed from the Transportation Network Providers (TNP) trip records released by the City of Chicago [6]. Both datasets provide origin–destination information, which we aggregate into region-level mobility graphs. For Chinese cities, mobility graphs are constructed by aggregating offline store-level transaction records from WeChat Pay (Tencent) into region-level interaction flows.

AlphaEarth embeddings. We use AlphaEarth embeddings as the geospatial foundation features in the main experiments. The embeddings are obtained from Google Earth Engine [13]. For each H3 region, we extract all 10-meter AlphaEarth embedding pixels within the region boundary and use their mean vector as the region-level representation.

C.2 Downstream Tasks

Shanghai and Guangzhou. For crime datasets, we aggregate the original crime cases data from Zhang et al. [47] over a 5-year interval. Night-time light labels are derived by aggregating VIIRS radiance pixels within each H3 region. We remove invalid and negative

pixels, compute region-level summary statistics, and use the log-transformed total radiance, $\log(1 + \text{sum})$, as the downstream NTL target [10]. Carbon emissions are derived from the ODIAC fossil-fuel CO₂ emission dataset [29]. We use monthly 1 km ODIAC 2024 GeoTIFF rasters for 2021, aggregate pixel values within each H3 region across all months, and use the log-transformed annual sum. PM_{2.5} concentration is derived from the CHAP / ChinaHighPM_{2.5} 1 km annual PM_{2.5} product [38, 39]. We use the 2021 NetCDF raster,

aggregate valid raster pixels within each H3 region by their mean value.

NYC and CHI. Crime records are collected from NYPD [28] and the Chicago Police Department [5]. Check-in activity is obtained from Foursquare [44] and house price labels are derived from Zillow [49] and NYC Department of Finance records [27]. We obtain regional income level data from the U.S. Census Bureau [36].